

4(a)	any four from: 1 manipulate DNA to modify organism's characteristics / AW ; 2 gene / allele / (section of) DNA, obtained using restriction, enzyme / endonuclease ; 3 use of reverse transcriptase to make, gene / allele / (section of) DNA, using mRNA ; 4 gene / allele / (section of) DNA, inserted into, vector / plasmid, using ligase ; 5 add / insert, (recombinant), vector / plasmid, into, (host) cell / bacterium ; 6 clone / multiply, cell / bacteria ; A put bacteria in a fermenter 7 gene is expressed and, protein / factor XIII / factor IX, is made ; 8 AVP ; e.g. artificial / chemical, synthesis of new gene	4
4(b)(i)	any three from: 1 haemophilia B (first) because (F9) gene is small enough to fit into virus / 1.6kbp and 4.7kbp ; ora 2 haemophilia A (first) because it is more common ; ora <i>therapy not attempted</i> 3 (because) disease can be managed by recombinant proteins ; 4 gene therapy (trials) involve, risk of harm / side effects / allergic reactions / immune response ; <i>therapy attempted</i> 5 provides a cure / AW ; 6 taking clotting factors etc. can involve risk of harm ;	3

Question	Answer	Marks
4(b)(ii)	any two from: 1 to allow binding of, RNA polymerase / transcription factors ; 2 to switch gene on / so gene is expressed / allow transcription (of gene) ; 3 at right time / all the time / in sufficient quantities ; 4 in correct tissue ;	2
4(b)(iii)	any two from: <i>decrease success because:</i> 1 antibodies / immune response, may, destroy / attack / AW, virus / vector ; 2 (so) limiting / stopping, delivery of, gene / allele / DNA, to cells ; 3 destroys GM cells or destroys cells that have successfully taken up the, virus / gene / allele / DNA ;	2
4(c)	any two from: <i>list rule</i> 1 (gene editing) is, precise / exact / accurate ; 2 patient's own gene can be, edited / corrected or no introduction of, gene / allele ; 3 no need to introduce promoter ; 4 less / no, risk of cancer or less / no, immune response ;	2